



epochs

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TEMPORAL EXPERIMENT

pre 1990-2009

post 2017-2026

fastText -> v_pre(sp)

fastText -> v_post(sp)

spatial drift =
haversine(centroid_pre,
centroid_post) km

embedding drift =
 $1 - \cos(v_pre, v_post)$

Spearman rho
(emb_drift, spatial_drift)

$\rho > 0$:
moved on map AND
in vector space
= *climate-tracking signal*

validate

PERMUTATION NULL

Pool pre+post,
shuffle epoch labels,
retrain

Repeat 10x -> null
distribution

Bombus:
observed rho = 0.38
null mean = 0.09
0/10 perms >= observed

SPATIAL TOKENISATION

Geohash bin (precision 4)
~20 x 20 km cells

Each cell = a 'sentence'
set of species observed
there
*distributional hypothesis
applied to ecology*

EMBEDDING (fastText skip-gram)

Predict a species'
neighbours
from the focal species

100-d vector per species
*compressed ecological
fingerprint*

GEOMETRIC PROPERTIES

$\cosine(sp_A, sp_B)$ - niche
overlap

Clusters recover
Wallace zoogeographic
realms

Nearest neighbours =
ecological analogs